Item 9: equal-advice audit and free-relation census

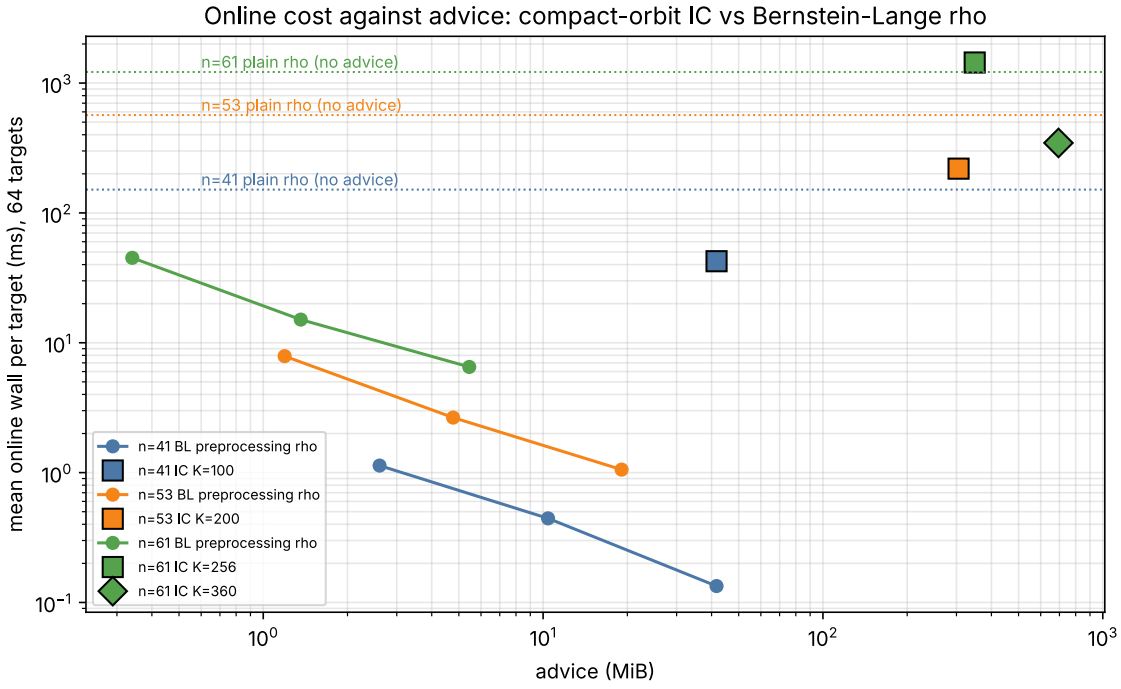
Item 9: the online IC wins at equal advice, and the free-relation census

Verdict: POSITIVE for the audit (Lead A). At the same advice size or less, Bernstein–Lange preprocessing rho solves a fresh Koblitz target 50–500 times faster online than compact-orbit index calculus. This holds on every rung that fits in 1 GiB (n = 41, 53 and 61). In every comparison the 95% interval lies far above 1.

The online-only IC “wins” (13.9x at n=61, 54.1x at N83) compare IC with a table against rho with no table. Once rho gets the same advice, they are not wins. Following the report’s kill criterion, this item recommends retiring them as wins from `docs/ic/LEADERBOARD.md` and landing no new online-only rungs.

Verdict: NEGATIVE for Lead B (the free-relation lever closes). The self-collision census matches the uniform E-3 law $K^4 n^3/(4r)$. The leads note’s prediction n^4K^4/r is too high by a factor of about 60. At n=61, K=600, there are 147 verified non-trivial collisions, but they span only **49 independent relations** (rank mod r), against 599 needed. That is below the 100-relation kill line, so the free-relation lever is closed at the rungs the repository runs.

Protocol: PROTOCOL.md, frozen before data. Raw rows: `data/`. Analysis: `scripts/analyze.py` → results.json.



online vs advice

Lead A: online cost at equal advice

All arms ran on the same 64 unseen public targets per rung (`data/targets_n*.txt`). Each target was checked against its held-back scalar (`data/scalars_n*.txt`): 64/64 for every IC and BL arm, and 64/64, 32/32 and 64/64 for plain rho at n=41, 53 and 61. Every arm ran single-threaded under `nice` and `prlimit --as=1 GiB`.

The unit is the per-target online wall time in ms. For IC this runs from the target query to the `[d]G = Q` check. For BL it covers the per-walk `[c]G`, the walk, the lookups and the check. Native counts are listed beside the times.

Ratios are IC/BL with a paired target bootstrap over 10,000 resamples.

rung	IC arm (advice)	IC online ms, mean / median	IC probes, mean	BL table (advice)	BL online ms, mean / median	BL steps, mean	IC / BL, ratio of means [95% CI]	median ratio [95% CI]
n=41	K=100 (43.7 MB)	42.5 / 32.5	19,328	2.73 M DPs (**43.7 MB, equal**)	0.134 / 0.063	102	**317 [137, 739]**	490 [360, 740]
n=53	K=200 (320 MB)	219 / 140	103,829	1.25 M DPs (20.0 MB, **1/16**)	1.06 / 0.23	646	**208 [103, 630]**	546 [297, 906]

rung	IC arm (advice)	IC online ms, mean / median	IC probes, mean	BL table (advice)	BL online ms, mean / median	BL steps, mean	IC / BL, ratio of means [95% CI]	median ratio [95% CI]
n=61	K=256 (366 MB)	1,440 / 830	384,349	357 k DPs (5.7 MB, **1/64**)	6.51 / 2.23	4,287	**221 [150, 327]**	196 [163, 254]
n=61	K=360 (728 MB)	346 / 222	168,331	same (5.7 MB, **1/128**)	6.51 / 2.23	4,287	**53 [35, 80]**	84 [34, 120]

The BL tables at n=53 and n=61 are smaller than IC's. The online walk is a deterministic function of the target and the seed, so a superset table can never take more steps on the same target. BL at equal bytes is therefore at least this fast. At n=41, where the table is exactly equal, BL is 317x faster. Even BL's smallest prefixes win at every rung: at 1/1,024 of IC's bytes at n=61, IC/BL is still 32 [22, 46] for K=256 and 7.7 [5.5, 10.6] for K=360 (`results.json`, `bl_level2`).

The native counts give the same answer without the clock. IC needs 20k–384k probes (one S3 root solve plus one table lookup each), while BL needs 100–4,300 point additions with canonicalisation. Per operation the two cost about the same: 4.1–7.5 µs per IC root call and 1.3–2.1 µs per BL step on this contended host.

Plain-rho reference (the published comparison, reproduced here). Strong rho (rung 3, 32 lanes, no table) against IC gives a rho/IC ratio of means of:

- n=41: 3.56 [2.66, 4.88];
- n=53: 3.61 [2.44, 5.57];
- n=61: 0.85 [0.63, 1.17] at K=256 and 3.52 [2.60, 4.87] at K=360.

So the IC arm shows the same kind of online "win" that the 13.9x row reports (K=500 there). Against the same rho given a table, BL beats plain rho by 187–1,130x (ratio of means). The table is what makes the difference, not index calculus.

Controls.

- *Positive (BL instrument law).* The log-log slope of BL online steps against table size over the three prefixes is −0.66 at n=41, −0.78 at n=53 and −0.65 at n=61. The law is −1/2 and the window was [−0.75, −0.25], so n=53 falls outside it by 0.03. Smaller prefixes reuse the DP density tuned for the full table (`2^t` is not re-optimised per prefix), which steepens the slope.
- *Null (identical map required).* At n=41 the table was built with one jump map and walked online with another (`bl_n41_mismatch`, 170 k DPs). Mean online steps were 32,670, against 626 for the matched map with the same number of entries: 52x worse. The cost reaches the random-hit level $N/T \approx 39{,}300$. The preregistered prediction "< 25% solved within 400·2^t steps" **failed**: 61/64 solved. The cause is the budget (102,400 steps), which was above N/T, so random DP hits finish the job. The effect the control was built to show, that the table helps only through the identical map, is clear.

Precompute. This is not part of the claim. It is shown so that the setup trade is visible:

rung	IC setup (index + rank stage), s	BL precompute, s (steps)
n=41	1.9 + 4.4	803 ($3.3 \cdot 10^8$)
n=53	8.0 + 41.3	852 ($1.2 \cdot 10^9$), 1/16 table
n=61	30.6 + 412.9 (K=256); 37.9 + 158.7 (K=360)	773 ($1.1 \cdot 10^9$), 1/64 table

All of these are wall times on a host with load average about 25 on 4 cores. The BL precompute walks are batched in 256 lanes and sit at about 230 ns per step of CPU time. About half of all DPs found were duplicates: the tables are saturated ($T \cdot W^2 \approx N$).

Projection to the published points (not measured)

These are computed from the fitted laws, not from runs.

- **n=61, K=500.** IC's root table alone is 1 GiB; the published mean IC online is not in this item. Scaling the K=360 IC online (346 ms) by $(360/500)^2$ gives about 180 ms. BL with the same ≈1.5 GB of advice has $\sqrt{(N/T)} \approx 120$ steps, about 260 steps at the measured 2.2x excess, which is under 1 ms. That projects IC/BL at a few hundred.
- **N83 K3240.** IC online is 58.9 ms (published mean). BL with 14 GiB holds $9.4 \cdot 10^8$ entries, so $\sqrt{(N/T)} \approx 230$ steps, which is about 1 ms. That projects IC/BL at about 50–100.

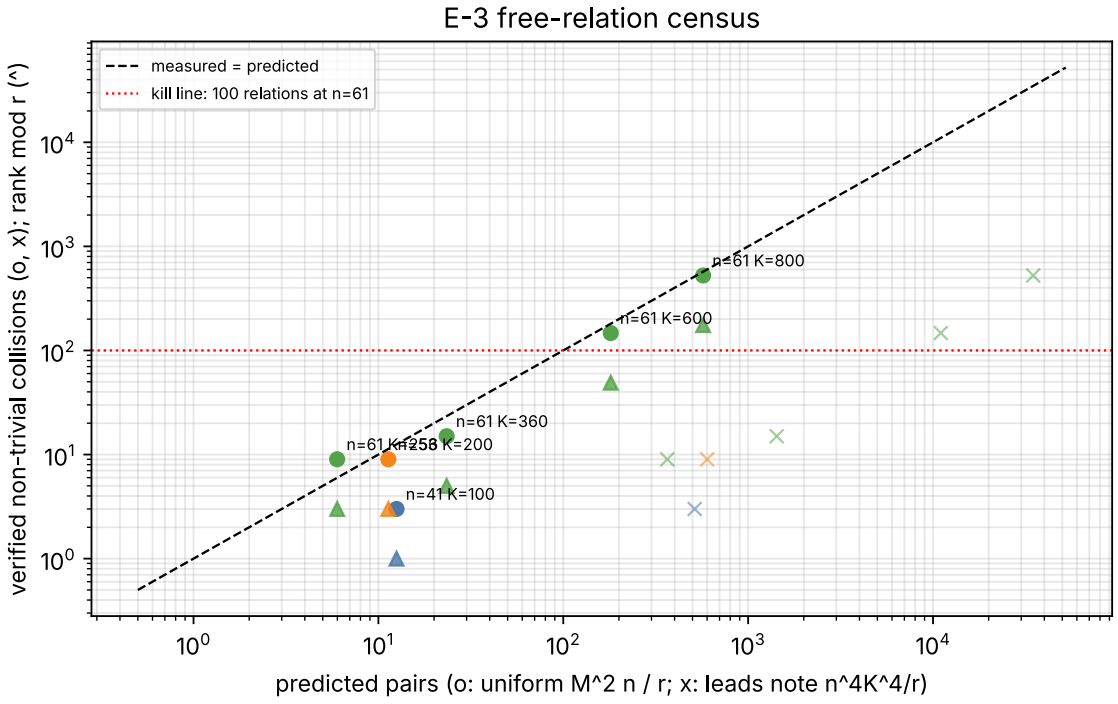
Neither the N83 point nor the n=61 K=500 point fits the 1 GiB cap, and the BL instrument is written for n ≤ 63 (single-word fields). These projections are not results.

Lead B: E-3 free-relation census

`ic_next10_09_equal_advice census` builds the same compact-orbit base as `construct:<n>:0:<K>` and enumerates every unordered pair state together with both S3 roots. It sorts the Frobenius-canonical roots and searches every equal key for a signed four-point identity whose log row is non-zero. It then computes the rank mod r of all verified rows. A collision is *trivial* only if every identity it carries has a zero row.

rung	K	root keys	collision pairs	trivial (= 2K)	verified non-trivial	uniform prediction $M^2 n/r$	leads note n^4K^4/r	rank mod r
n=41	100	0.41 M	203	200	3	12.5	514	1
n=53	200	2.1 M	409	400	9	11.3	600	3
n=61	256	4.0 M	521	512	9	6.0	365	3

rung	K	root keys	collision pairs	trivial (= 2K)	verified non-trivial	uniform prediction M^2n/r	leads note n^4K^4/r	rank mod r
n=61	360	7.9 M	735	720	15	23.4	1,428	5
n=61	**600**	22.0 M	1,347	1,200	**147**	180.6	11,016	**49**
n=61	800	39.0 M	2,125	1,600	525	570.7	34,817	175



census

- Every collision verified as a group identity, with 0 verification failures.
- The non-trivial count follows the uniform law, measured/predicted 0.24–1.5 (0.81 at K=600 and 0.92 at K=800). The leads note is wrong by a factor of n.
- Exactly 2K collisions per table are structural trivial ones: the same pair orbit is reached twice, so the row is zero.
- The rank is exactly one third of the non-trivial count in every row (147 → 49, 525 → 175). The likeliest reason is that each four-point relation `P_a+P_b+P_c+P_d = 0` shows up once for each of its three pairings, but this was not separately checked. Distinct relations therefore come to ≈ $K^4n^3/(12r)$. Full rank needs that to reach K, which means $K \approx (12r/n^3)^{1/3} \approx 2{,}050$ at n=61: a $3.4 \cdot 10^8$-state table, more than 10x the K=600 one.
- **Kill criterion:** 49 independent relations at n=61, K=600 (147 raw verified collisions), against the 100 line. The lever is closed.

Deviations from the protocol

1. **Advice scaled down at n=53 and n=61** (BL at 1/16 and 1/64 of IC's bytes instead of equal and 1/4). The cause was host contention: load average 25–27 on 4 cores from the other nine items, which left this item about 13–25% of a core. A 1/4-bytes n=61 precompute was projected to pass the 45-minute job cap, and two attempts (1/4, then 1/16) were killed before finishing, so no data came from them. Because of per-target monotonicity in table size, this makes the comparison more favourable to IC, not less.
2. The null control used 1/16 bytes (it does not need equal advice), and its preregistered success-rate prediction failed (see above).
3. The n=53 plain-rho reference used 32 of the 64 targets, as the run script set it up before launch. This is a reference arm, not the primary comparison.
4. The census instrument was first built returning the first identity found. The rebuilt version prefers a non-trivial identity, and the counts above come from it. The n=61 K=600 figures came out identical in both versions.
5. **Wall-clock noise.** The jobs ran under heavy, variable contention. IC K=256 ran in the busiest window (7.5 µs per root call, against 4.1 µs at K=360). The per-target pairing does not remove that. The 50–500x gaps are far larger than this noise, and the native counts agree.
6. There are no hardware counters (`perf_event_open` is unavailable), so the count unit is native operations, not instructions.

Reproduce

```
        cargo build --release --example ic_next10_09_equal_advice \
          --example koblitz_orbit_dlp_fast_online --example koblitz_rho_batch_ks_strong_online
        B=target/release/examples; D=research/ic_next10_20261014/09-equal-advice/data
        for n in 41 53 61; do $B/ic_next10_09_equal_advice targets $n 0 64 eval-v1
$D/targets_n$n.txt $D/scalars_n$n.txt; done
        # IC arm
```

```
        $B/koblitz_orbit_dlp_fast_online construct:61:0:256 $D/targets_n61.txt 7
$D/ic_n61_K256.jsonl
        # BL arm (bytes = IC formula / 64; scripts/ic_bytes.py 256 61)
        $B/ic_next10_09_equal_advice bl 61 0 5711008 $D/targets_n61.txt $D/bl_n61_64th.jsonl
        # null control
        BL_ONLINE_MAP_SEED=mismatch BL_PREFIXES=1 $B/ic_next10_09_equal_advice bl 41 0 2730602
$D/targets_n41.txt $D/bl_n41_mismatch.jsonl
        # plain rho, one process per target
        KIC_RHO_TARGET_POINT=x,y KIC_RHO_RUNG=3 KIC_RHO_LANES=32 KIC_RHO_DP_BITS=4
$B/koblitz_rho_batch_ks_strong_online 61 0 signed_frobenius 1 610000
        # census
        $B/ic_next10_09_equal_advice census 61 0 600 $D/census_n61_K600.jsonl
        python3 -I research/ic_next10_20261014/09-equal-advice/scripts/analyze.py
```

The full job list is in `scripts/run_stream.sh` (streams `s1`, `s2`), and the timestamps are in `data/jobs.log`. Every job ran under `nice`, `prlimit --as=1073741824` and `RAYON_NUM_THREADS=1`.

Files

- `examples/ic_next10_09_equal_advice.rs`: new example with the `targets`, `bl` and `census` modes (no `src/` edits).
- `PROTOCOL.md`, `README.md`, `README.pdf`, `results.json`.
- `fig_online_vs_advice.{svg,png}` and `fig_census.{svg,png}`.
- `scripts/ic_bytes.py`, `scripts/run_stream.sh`, `scripts/analyze.py`.
- `data/`: the targets and scalars, IC/BL/rho/census rows, and `jobs.log`.